

MOHAN KUMAR K

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PROFESSIONAL SUMMARY

Computer Science graduate (M.E. pursuing) with hands-on experience in Full-stack web development and machine learning. Delivered 2 end-to-end ML projects using Python, Django, TensorFlow, and OpenCV. Seeking a Software Developer Intern role to contribute strong problem-solving and engineering skills in a fast-paced team.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: Django, Bootstrap, TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn, Pandas, NumPy, Langchain, HuggingFace, MediaPipe

ML / AI: CNNs, NLP, Deep Learning, Computer Vision, Image Processing

Tools: Git, GitHub, VS Code, REST APIs

EXPERIENCE

Retech Solution — Python & Django Intern

May 2023 – Jul 2023

Full-Stack Web Development Internship

- Engineered a Django web application, reducing manual task execution time by ~40% through Python-driven automation of repetitive backend workflows.
- Built RESTful backend modules for user authentication, session management, and data processing, delivering production-ready features ahead of schedule.
- Collaborated in an agile team of 4, consistently meeting sprint milestones and improving overall team delivery speed.

PROJECTS

Fish Species Identification & Freshness Detection System

2026 – 2027

Python • TensorFlow • Keras • OpenCV • CNN • Scikit-learn

- Designed and trained a CNN model achieving high-accuracy classification of fish species from image datasets using automated feature extraction.
- Integrated image preprocessing pipeline (resizing, normalization, noise removal) to improve model accuracy by optimizing input data quality.
- Developed freshness detection module by analyzing visual features — gills, eyes, skin texture — enabling rapid, non-invasive quality assessment.
- Built a real-time web interface for image upload, inference, and result visualization, reducing manual inspection effort for end users.

Real-Time Sign Language Translation System

2023 – 2024

Python • OpenCV • TensorFlow • MediaPipe • CNN • NLP • Pandas • NumPy

- Architected a real-time gesture recognition system translating sign language into text and speech, achieving live video inference with low latency.
- Implemented CNN-based gesture classifier leveraging MediaPipe landmark detection, reducing misclassification rate by applying data augmentation.
- Integrated an NLP post-processing layer to improve semantic coherence of sign-to-text output, increasing readability by 30%.
- Processed and visualized training metrics with Pandas & Matplotlib, enabling data-driven model tuning across multiple training cycles.

EDUCATION

Easwari Engineering College — M.E. Computer Science Engineering | CGPA: 8.7/10

2025 – 2027 (Pursuing)

DMI College of Engineering — B.E. Computer Science Engineering | CGPA: 7.7/10

2020 – 2024

CERTIFICATIONS

- Udemy — Python Web Development (Django, SQL, REST APIs, HTML, CSS, JavaScript)
- Udemy — JavaScript | Udemy — SQL
- DMI College — Full-Stack Web Development Workshop (Responsive Application Development)